

Corey Oses

Materials Science and Engineering, Johns Hopkins University

email corey@jhu.edu
phone (M) +1 (201) 674 1407 · (W) +1 (410) 516 5779
website coreyoses.com
Google Scholar [user=Za7m4CMAAAAJ](https://scholar.google.com/citations?user=Za7m4CMAAAAJ) · citations: 10121 (YTD 1307) · h-index: 36

Work Experience

Assistant Professor	2022–present	Johns Hopkins University
Postdoctoral Fellow	2018–2022	Duke University
Internship	Summer 2013	Cornell High Energy Synchrotron Source (BioSAXS on F2 and G Beamlines)
Internship	Summer 2012	Cornell High Energy Synchrotron Source (Capillary Optics Group)

Education

Ph.D.	2013–2018	Duke University Department: Mechanical Engineering and Materials Science Thesis: <i>Machine learning, phase stability, and disorder with the Automatic Flow Framework for Materials Discovery</i> DukeSpace: hdl.handle.net/10161/18254
B.Sc.	2009–2013	Cornell University Department: Applied and Engineering Physics Thesis: <i>Plume Propagation Simulation for Pulsed Laser Deposition</i>

Funding

Current

7. **Title:** *Addressing the Cesium Problem for Negative Hydrogen Ion Sources*
Duration: 04/01/2026 – 03/31/2028
Funding Level: \$500,000
Agency: Advanced Research Projects Agency-Energy (ARPA-E), Inspiring Generations of New Innovators to Impact Technologies in Energy 2025 (IGNITE 2025)
Institution: Johns Hopkins University
PI: C. Oses
6. **Title:** *Multi-Principal Element Alloys for Extreme Environments*
Duration: 03/01/2026 – 02/28/2027
Funding Level: \$262,675
Agency: Department of the Air Force, AFWERX SBIR
Prime: Avalanche Inc.
Subaward Institution: Johns Hopkins University
Subaward PI: C. Oses
5. **Title:** *Next-Generation Mirrors for Harsh Fusion Environments*
Duration: 01/01/2026 – 12/31/2028
Funding Level: \$450,000
Agency: Seaver Institute
Institution: Johns Hopkins University
PI: C. Oses
4. **Title:** *From Laboratory to Launchpad: Transition Pathways for Climate Science and Energy IP*
Duration: 07/01/2025 – 06/30/2026
Funding Level: \$115,000
Agency: Johns Hopkins University, Nexus Convening Award
Institutions: Johns Hopkins University Applied Physics Laboratory, Johns Hopkins University, and Johns Hopkins Carey Business School
PIs: A. G. Bregman & C. Oses; **co-PIs:** S. D. Cohen & J. Erlebacher
3. **Title:** *AI-Driven Discovery of High-Entropy H_2 Generators (ADD- H_2)*
Duration: 07/11/2024 – 06/30/2027
Funding Level: \$2,114,366
Agency: Johns Hopkins University, SURPASS
Institutions: Johns Hopkins University and Johns Hopkins University Applied Physics Laboratory
PIs: C. Oses & A. G. Bregman; **co-PIs:** K. A. Kane, K. K. Rao, V. Leon, A. S. Hall & L. H. Hamilton
2. **Title:** *High-Entropy Glass-Ceramics for Nuclear Waste Immobilization*
Duration: 09/21/2023 – 03/19/2027
Funding Level: \$657,945
Agency: Advanced Research Projects Agency-Energy (ARPA-E), Creating Revolutionary Energy And Technology Endeavors (CREATE) 2023
Institution: Johns Hopkins University
PI: C. Oses
1. **Title:** *Startup Package*
Duration: 07/01/2022 – 06/30/2024
Funding Level: \$664,000
Agency: Johns Hopkins University
Institution: Johns Hopkins University
PI: C. Oses

Pending

2. **Title:** *C-DRIVE: Catalyst Design and Deployment Combining Robotics, Artificial Intelligence, Reactor Validation, and Entropy Engineering*
Duration: 07/01/2026 – 06/30/2029
Funding Level: \$3,500,000; **Cost-Share:** \$3,888,888.89
Agency: Advanced Research Projects Agency-Energy (ARPA-E), Catalytic Application Testing for Accelerated Learning Chemistries via High-Throughput Experimentation and Modeling Efficiently (CATALCHEM-E)
Institutions: Johns Hopkins University, Massachusetts Institute of Technology, and GE Vernova Advanced Research
PI: C. Oses; **co-PIs:** J. Li & N. Thiagarajan

1. **Title:** *Rapid, AI-enhanced Magnetic Material Discovery (RAMMED)*
Duration: 05/01/2026 – 04/30/2029
Funding Level: \$3,000,000; **Cost-Share:** \$373,000
Agency: Advanced Research Projects Agency-Energy (ARPA-E), Magnetic Acceleration Generating New Innovations and Tactical Outcomes (MAGNITO)
Institutions: Johns Hopkins University, University of Maryland Department of Materials Science and Engineering, Ames National Laboratory, and GE Vernova Advanced Research
PI: C. Oses; **co-PIs:** I. Takeuchi, J. Cui & J. R. Owens

Past

7. **Title:** *Building Trusted Thermodynamic Datasets for Materials Discovery of Metal Iodides*
Duration: 03/01/2025 – 09/30/2025
Funding Level: \$25,000
Agency: Data Science and AI Institute, 2025 Trusted Dataset Award
Institution: Johns Hopkins University
PI: C. Oses
6. **Title:** *Accelerated Materials Discovery for High-Entropy Hydrogen Fuel Cell Catalysts (HE-FCC)*
Duration: 06/05/2024 – 08/13/2024
Funding Level: \$6,000
Agency: Institute for Data Intensive Engineering and Science, 2024 Summer Student Fellowship
Institution: Johns Hopkins University
PI: C. Oses
5. **Title:** *Waste-Heat Powered Hydrogen Production on Mars*
Duration: 09/01/2023 – 08/31/2024
Funding Level: \$25,000
Agency: Space@Hopkins, 2023 Seed Funding Program
Institutions: Johns Hopkins University and Johns Hopkins University Applied Physics Laboratory
PI: C. Oses; **co-PIs:** K. A. Kane & A. G. Bregman
4. **Title:** *High-Entropy Capacitive Energy Storage*
Duration: 07/01/2023 – 06/30/2024
Funding Level: \$150,000
Agency: Johns Hopkins University, Discovery Award
Institution: Johns Hopkins University
PI: C. Oses; **co-PI:** S. M. Koohpayeh
3. **Title:** *AFLUX@JHU: Materials Search-API for the JHU aflow.org Data Repositories*
Duration: 06/05/2023 – 08/13/2023
Funding Level: \$6,000
Agency: Institute for Data Intensive Engineering and Science, 2023 Summer Student Fellowship
Institution: Johns Hopkins University
PI: C. Oses
2. **Title:** *Accelerated Disordered Materials Discovery for Energy Generation in Space*
Duration: 06/05/2023 – 08/13/2023
Funding Level: \$18,796; **Cost-Share:** \$10,996
Agency: Maryland Space Grant Consortium, 2023 Summer Student Internship Program
Institution: Johns Hopkins University
PI: C. Oses
1. **Title:** *High-Entropy Metal-Organic Frameworks for High-Performance Lithium-Sulfur Batteries*
Duration: 04/01/2023 – 03/31/2024
Funding Level: \$25,000
Agency: Institute for Data Intensive Engineering and Science, 2023 Seed Funding Initiative
Institution: Johns Hopkins University
PI: C. Oses; **co-PI:** V. S. Thoi

Journal Publications [†] contributed equally * corresponding

2026

49. X. Xu, G. Han, T. Li, J. Hu, G. Qiu, S.-Y. Tseng, G. H. Han, A. Dupuy, A. S. Hall & C. Oses*, *Stabilizing Iodine in Pyrochlore: Toward New Nuclear Waste Forms*, *Precis. Chem.* (2026). DOI: [10.1021/prechem.5c00467](https://doi.org/10.1021/prechem.5c00467). [PDF]

2025

48. G. Han[†], T. Li[†], X. Xu, J. Lee, S. Sequeira, A. Ajith & C. Oses*, *The search for high-entropy fuel-cell catalysts using disorder descriptors*, *Nano Futures* **9**, 045001 (2025). DOI: [10.1088/2399-1984/ae19b0](https://doi.org/10.1088/2399-1984/ae19b0). [PDF]

47. C. Oses*, T. Li, X. Xu, G. Han, G. Qiu & J. R. Owens, *Beyond the four core effects: revisiting thermoelectrics with a high-entropy design*, Mater. Horiz. **12**, 5946–5956 (2025). DOI: [10.1039/D5MH00356C](https://doi.org/10.1039/D5MH00356C). [PDF]

- This paper was selected to feature in **2025 Most Popular Articles** by the Royal Society of Chemistry: Materials Horizons (2026).

46. G. Qiu, T. Li, X. Xu, Y. Liu, M. Niyogi, K. Cariaga & C. Oses*, *High entropy powering green energy: hydrogen, batteries, electronics, and catalysis*, npj Comput. Mater. **11**, 145 (2025). DOI: [10.1038/s41524-025-01594-6](https://doi.org/10.1038/s41524-025-01594-6). [PDF]

2024

45. T. Gong, G. Qiu, M.-R. He, O. V. Safonova, W.-C. Yang, D. Raciti, C. Oses & A. S. Hall, *Atomic Ordering-Induced Ensemble Variation in Alloys Governs Electrocatalyst On/Off States*, J. Am. Chem. Soc. **147**(1), 510–518 (2024). DOI: [10.1021/jacs.4c11753](https://doi.org/10.1021/jacs.4c11753). [PDF]
44. K. S. Vecchio, S. Curtarolo, K. Kaufmann, T. J. Harrington, C. Oses & C. Toher, *Fermi energy engineering of enhanced plasticity in high-entropy carbides*, Acta Mater. **276**, 120117 (2024). DOI: [10.1016/j.actamat.2024.120117](https://doi.org/10.1016/j.actamat.2024.120117). [PDF]
43. M. L. Evans, J. Bergsma, A. Merkys, C. W. Andersen, O. B. Andersson, D. Beltrán, E. Blokhin, T. M. Bolland, R. Castañeda Balderas, K. Choudhary, A. Díaz Díaz, R. Domínguez García, H. Eckert, K. Eimre, M. E. Fuentes-Montero, A. M. Krajewski, J. J. Mortensen, J. M. Nápoles-Duarte, J. Pietryga, J. Qi, F. d. J. Trejo Carrillo, A. Vaitkus, J. Yu, A. C. Zettl, P. Baptista de Castro, J. Carlsson, T. F. T. Cerqueira, S. Divilov, H. Hajiyani, F. Hanke, K. Jose, C. Oses, J. Riebesell, J. Schmidt, D. Winston, C. Xie, X. Yang, S. Bonella, S. Botti, S. Curtarolo, C. Draxl, L. E. Fuentes Cobas, A. Hospital, Z.-K. Liu, M. A. L. Marques, N. Marzari, A. J. Morris, S. P. Ong, M. Orozco, K. A. Persson, K. S. Thygesen, C. Wolverton, M. Scheidgen, C. Toher, G. J. Conduit, G. Pizzi, S. Gražulis, G.-M. Rignanese & R. Armiento, *Developments and applications of the OPTIMADE API for materials discovery, design, and data exchange*, Digit. Discov. **3**, 1509–1533 (2024). DOI: [10.1039/D4DD00039K](https://doi.org/10.1039/D4DD00039K). [PDF]
42. S. Divilov†, H. Eckert†, D. Hicks†, C. Oses†, C. Toher†, R. Friedrich, M. Esters, M. J. Mehl, A. C. Zettl, Y. Lederer, E. Zurek, J.-P. Maria, D. W. Brenner, X. Campilongo, S. Filipović, W. G. Fahrenholtz, C. J. Ryan, C. M. DeSalle, R. J. Creales, D. E. Wolfe, A. Calzolari & S. Curtarolo, *Disordered enthalpy-entropy descriptor for high-entropy ceramics discovery*, Nature **625**, 66–73 (2024). DOI: [10.1038/s41586-023-06786-y](https://doi.org/10.1038/s41586-023-06786-y). [PDF]
41. A. B. Peters, D. Zhang, S. Chen, C. Ott, C. Oses, S. Curtarolo, I. McCue, T. Pollock & S. E. Prameela, *Materials Design for Hypersonics*, Nat. Commun. **15**, 3328 (2024). DOI: [10.1038/s41467-024-46753-3](https://doi.org/10.1038/s41467-024-46753-3). [PDF]
- This paper was selected for **Editors' Highlight** by Springer Nature (2024).

2023

40. D. E. Wolfe, C. M. DeSalle, C. J. Ryan, R. E. Slapikas, R. T. Sweny, R. J. Creales, P. A. Kolonin, S. P. Stepanoff, A. Haque, S. Divilov, H. Eckert, C. Oses, M. Esters, D. W. Brenner, W. G. Fahrenholtz, J.-P. Maria, C. Toher, E. Zurek & S. Curtarolo, *Influence of Processing on the Microstructural Evolution and Multiscale Hardness in Titanium Carbonitrides (TiCN) Produced via Field Assisted Sintering Technology*, Materialia **27**, 101682 (2023). DOI: [10.1016/j.mtl.2023.101682](https://doi.org/10.1016/j.mtl.2023.101682). [PDF]
39. C. Oses, M. Esters, D. Hicks, S. Divilov, H. Eckert, R. Friedrich, M. J. Mehl, A. Smolyanyuk, X. Campilongo, A. van de Walle, J. Schroers, A. G. Kusne, I. Takeuchi, E. Zurek, M. Buongiorno Nardelli, M. Fornari, Y. Lederer, O. Levy, C. Toher & S. Curtarolo, *aflow++: a C++ framework for autonomous materials design*, Comput. Mater. Sci. **217**, 111889 (2023). DOI: [10.1016/j.commatsci.2022.111889](https://doi.org/10.1016/j.commatsci.2022.111889). [PDF]
- This paper was selected for **Editor's Choice** by Elsevier (2022).
38. M. Esters, C. Oses, S. Divilov, H. Eckert, R. Friedrich, D. Hicks, M. J. Mehl, F. Rose, A. Smolyanyuk, A. Calzolari, X. Campilongo, C. Toher & S. Curtarolo, *aflow.org: a web ecosystem of databases, software and tools*, Comput. Mater. Sci. **216**, 111808 (2023). DOI: [10.1016/j.commatsci.2022.111808](https://doi.org/10.1016/j.commatsci.2022.111808). [PDF]
37. M. Esters†, A. Smolyanyuk†, C. Oses, D. Hicks, S. Divilov, H. Eckert, X. Campilongo, C. Toher & S. Curtarolo, *QH-POCC: taming tiling entropy in thermal expansion calculations of disordered materials*, Acta Mater. **245**, 118594 (2023). DOI: [10.1016/j.actamat.2022.118594](https://doi.org/10.1016/j.actamat.2022.118594). [PDF]

2022

36. A. Calzolari, C. Oses, C. Toher, M. Esters, X. Campilongo, S. P. Stepanoff, D. E. Wolfe & S. Curtarolo, *Plasmonic high-entropy carbides*, Nat. Commun. **13**, 5993 (2022). DOI: [10.1038/s41467-022-33497-1](https://doi.org/10.1038/s41467-022-33497-1). [PDF]
35. X. Wang, D. M. Proserpio, C. Oses, C. Toher, S. Curtarolo & E. Zurek, *The Microscopic Diamond Anvil Cell: Stabilization of Superhard, Superconducting Carbon Allotropes at Ambient Pressure*, Angew. Chem. **61**(32), e202205129 (2022). DOI: [10.1002/anie.202205129](https://doi.org/10.1002/anie.202205129). [PDF]
34. H. J. Kulik, T. Hammerschmidt, J. Schmidt, S. Botti, M. A. L. Marques, M. Boley, M. Scheffler, M. Todorović, P. Rinke, C. Oses, A. Smolyanyuk, S. Curtarolo, A. Tkatchenko, A. P. Bartók, S. Manzhos, M. Ihara, T. Carrington, J. Behler, O. Isayev, M. Veit, A. Grisafi, J. Nigam, M. Ceriotti, K. T. Schütt, J. Westermayr, M. Gastegger, R. J. Maurer, B. Kalita, K. Burke, R. Nagai, R. Akashi, O. Sugino, J. Hermann, F. Noé, S. Pilati, C. Draxl, M. Kuban, S. Rigamonti, M. Scheidgen, M. Esters, D. Hicks, C. Toher, P. V. Balachandran, I. Tamblyn, S. Whitlam, C. Bellinger & L. M. Ghiringhelli, *Roadmap on Machine Learning in Electronic Structure*, Electron. Struct. **4**(2), 023004 (2022). DOI: [10.1088/2516-1075/ac572f](https://doi.org/10.1088/2516-1075/ac572f). [PDF]
33. A. G. Kusne, A. McDannald, B. DeCost, C. Oses, C. Toher, S. Curtarolo, A. Mehta & I. Takeuchi, *Physics in the Machine: Integrating Physical Knowledge in Autonomous Phase-Mapping*, Front. Phys. **10**, 815863 (2022). DOI: [10.3389/fphy.2022.815863](https://doi.org/10.3389/fphy.2022.815863). [PDF]
32. C. Toher, C. Oses, M. Esters, D. Hicks, G. N. Kotsonis, C. M. Rost, D. W. Brenner, J.-P. Maria & S. Curtarolo, *High-entropy ceramics: Propelling applications through disorder*, MRS Bull. **47**, 194–202 (2022). DOI: [10.1557/s43577-022-00281-x](https://doi.org/10.1557/s43577-022-00281-x). [PDF]

2021

31. M. Esters, C. Oses, D. Hicks, M. J. Mehl, M. Jahnátek, M. D. Hossain, J.-P. Maria, D. W. Brenner, C. Toher & S. Curtarolo, *Settling the matter of the role of vibrations in the stability of high-entropy carbides*, Nat. Commun. **12**, 5747 (2021). DOI: [10.1038/s41467-021-25979-5](https://doi.org/10.1038/s41467-021-25979-5). [PDF]
 - This paper was selected for **Editors' Highlight** by Springer Nature (2021).
30. M. D. Hossain, T. Borman, C. Oses, M. Esters, C. Toher, L. Feng, A. Kumar, W. G. Fahrenholtz, S. Curtarolo, D. W. Brenner, J. M. LeBeau & J.-P. Maria, *Entropy Landscaping of High-Entropy Carbides*, Adv. Mater. **33**(42), 2102904 (2021). DOI: [10.1002/adma.202102904](https://doi.org/10.1002/adma.202102904). [PDF]
29. C. W. Andersen[†], R. Armiento[†], E. Blokhin[†], G. J. Conduit[†], S. Dwaraknath[†], M. L. Evans[†], Á. Fekete[†], A. Gopakumar[†], S. Gražulis[†], A. Merkys[†], F. Mohamed[†], C. Oses[†], G. Pizzi[†], G.-M. Rignanese[†], M. Scheidgen[†], L. Talirz[†], C. Toher[†], D. Winston[†], R. Aversa, K. Choudhary, P. Colinet, S. Curtarolo, D. Di Stefano, C. Draxl, S. Er, M. Esters, M. Fornari, M. Giantomassi, M. Govoni, G. Hautier, V. Hegde, M. K. Horton, P. Huck, G. Huhs, J. Hummelshøj, A. Kariryaa, B. Kozinsky, S. Kumbhar, M. Liu, N. Marzari, A. J. Morris, A. Mostofi, K. A. Persson, G. Petretto, T. Purcell, F. Ricci, F. Rose, M. Scheffler, D. Speckhard, M. Uhrin, A. Vaitkus, P. Villars, D. Waroquiers, C. Wolverton, M. Wu & X. Yang, *OPTIMADE: an API for exchanging materials data*, Sci. Data **8**, 217 (2021). DOI: [10.1038/s41597-021-00974-z](https://doi.org/10.1038/s41597-021-00974-z). [PDF]
28. R. Friedrich, M. Esters, C. Oses, S. Ki, M. J. Brenner, D. Hicks, M. J. Mehl, C. Toher & S. Curtarolo, *Automated coordination corrected enthalpies with AFLOW-CCE*, Phys. Rev. Mater. **5**, 043803 (2021). DOI: [10.1103/PhysRevMaterials.5.043803](https://doi.org/10.1103/PhysRevMaterials.5.043803). [PDF]
27. D. Hicks, M. J. Mehl, M. Esters, C. Oses, O. Levy, G. L. W. Hart, C. Toher & S. Curtarolo, *The AFLOW Library of Crystallographic Prototypes: Part 3*, Comput. Mater. Sci. **199**, 110450 (2021). DOI: [10.1016/j.commatsci.2021.110450](https://doi.org/10.1016/j.commatsci.2021.110450).
26. M. J. Mehl, M. Ronquillo, D. Hicks, M. Esters, C. Oses, R. Friedrich, A. Smolyanyuk, E. Gossett, D. Finkenshtadt & S. Curtarolo, *Tin-pest problem as a test of density functionals using high-throughput calculations*, Phys. Rev. Mater. **5**, 083608 (2021). DOI: [10.1103/PhysRevMaterials.5.083608](https://doi.org/10.1103/PhysRevMaterials.5.083608). [PDF]
25. M. D. Hossain[†], T. Borman[†], A. Kumar, X. Chen, A. Khosravani, S. R. Kalidindi, E. A. Paisley, M. Esters, C. Oses, C. Toher, S. Curtarolo, J. M. LeBeau, D. W. Brenner & J.-P. Maria, *Carbon Stoichiometry and Mechanical Properties of High Entropy Carbides*, Acta Mater. **215**, 117051 (2021). DOI: [10.1016/j.actamat.2021.117051](https://doi.org/10.1016/j.actamat.2021.117051). [PDF]

2020

24. A. G. Kusne[†], H. Yu[†], C. Wu, H. Zhang, J. Hattrick-Simpers, B. DeCost, S. Sarker, C. Oses, C. Toher, S. Curtarolo, A. V. Davydov, R. Agarwal, L. A. Bendersky, M. Li, A. Mehta & I. Takeuchi, *On-the-fly Closed-loop Autonomous Materials Discovery via Bayesian Active Learning*, Nat. Commun. **11**, 5966 (2020). DOI: [10.1038/s41467-020-19597-w](https://doi.org/10.1038/s41467-020-19597-w). [PDF]
23. K. Kaufmann, D. Maryanovsky, W. M. Mellor, C. Zhu, A. S. Rosengarten, T. J. Harrington, C. Oses, C. Toher, S. Curtarolo & K. S. Vecchio, *Discovery of novel high-entropy ceramics via machine learning*, npj Comput. Mater. **6**, 42 (2020). DOI: [10.1038/s41524-020-0317-6](https://doi.org/10.1038/s41524-020-0317-6). [PDF]
22. C. Oses, C. Toher & S. Curtarolo, *High-entropy ceramics*, Nat. Rev. Mater. **5**, 295–309 (2020). DOI: [10.1038/s41578-019-0170-8](https://doi.org/10.1038/s41578-019-0170-8). [PDF]
 - This paper was highlighted as a “**hot paper**” by Web of Science (Clarivate Analytics) (Nov 16, 2021).

2019

21. D. C. Ford, D. Hicks, C. Oses, C. Toher & S. Curtarolo, *Metallic glasses for biodegradable implants*, Acta Mater. **176**, 297–305 (2019). DOI: [10.1016/j.actamat.2019.07.008](https://doi.org/10.1016/j.actamat.2019.07.008). [PDF]
20. P. Avery, X. Wang, C. Oses, E. Gossett, D. M. Proserpio, C. Toher, S. Curtarolo & E. Zurek, *Predicting Superhard Materials via a Machine Learning Informed Evolutionary Structure Search*, npj Comput. Mater. **5**, 89 (2019). DOI: [10.1038/s41524-019-0226-8](https://doi.org/10.1038/s41524-019-0226-8). [PDF]
19. C. Toher, C. Oses, D. Hicks & S. Curtarolo, *Unavoidable disorder and entropy in multi-component systems*, npj Comput. Mater. **5**, 69 (2019). DOI: [10.1038/s41524-019-0206-z](https://doi.org/10.1038/s41524-019-0206-z). [PDF]
18. R. Friedrich, D. Usanmaz, C. Oses, A. R. Supka, M. Fornari, M. Buongiorno Nardelli, C. Toher & S. Curtarolo, *Coordination corrected ab initio formation enthalpies*, npj Comput. Mater. **5**, 59 (2019). DOI: [10.1038/s41524-019-0192-1](https://doi.org/10.1038/s41524-019-0192-1). [PDF]
17. P. Nath, D. Usanmaz, D. Hicks, C. Oses, M. Fornari, M. Buongiorno Nardelli, C. Toher & S. Curtarolo, *AFLOW-QHA3P: Robust and automated method to compute thermodynamic properties of solids*, Phys. Rev. Mater. **3**, 073801 (2019). DOI: [10.1103/PhysRevMaterials.3.073801](https://doi.org/10.1103/PhysRevMaterials.3.073801). [PDF]

2018

16. C. Oses, E. Gossett, D. Hicks, F. Rose, M. J. Mehl, E. Perim, I. Takeuchi, S. Sanvito, M. Scheffler, Y. Lederer, O. Levy, C. Toher & S. Curtarolo, *AFLOW-CHULL: Cloud-oriented platform for autonomous phase stability analysis*, J. Chem. Inf. Model. **58**(12), 2477–2490 (2018). DOI: [10.1021/acs.jcim.8b00393](https://doi.org/10.1021/acs.jcim.8b00393). [PDF]
15. C. Oses, C. Toher & S. Curtarolo, *Data-driven design of inorganic materials with the Automatic Flow Framework for Materials Discovery*, MRS Bull. **43**(9), 670–675 (2018). DOI: [10.1557/mrs.2018.207](https://doi.org/10.1557/mrs.2018.207). [PDF]
14. P. Sarker[†], T. J. Harrington[†], C. Toher, C. Oses, M. Samiee, J.-P. Maria, D. W. Brenner, K. S. Vecchio & S. Curtarolo, *High-entropy high-hardness metal carbides discovered by entropy descriptors*, Nat. Commun. **9**, 4980 (2018). DOI: [10.1038/s41467-018-07160-7](https://doi.org/10.1038/s41467-018-07160-7). [PDF]
13. V. Stanev, C. Oses, A. G. Kusne, E. Rodriguez, J. Paglione, S. Curtarolo & I. Takeuchi, *Machine learning modeling of superconducting critical temperature*, npj Comput. Mater. **4**, 29 (2018). DOI: [10.1038/s41524-018-0085-8](https://doi.org/10.1038/s41524-018-0085-8). [PDF]

12. E. Gossett, C. Toher, **C. Oses**, O. Isayev, F. Legrain, F. Rose, E. Zurek, J. Carrete, N. Mingo, A. Tropsha & S. Curtarolo, *AFLOW-ML: A RESTful API for machine-learning prediction of materials properties*, *Comput. Mater. Sci.* **152**, 134–145 (2018). DOI: [10.1016/j.commatsci.2018.03.075](https://doi.org/10.1016/j.commatsci.2018.03.075). [PDF]
 - This paper was selected for **Editor's Choice** by Elsevier (2018).
11. D. Hicks, **C. Oses**, E. Gossett, G. Gomez, R. H. Taylor, C. Toher, M. J. Mehl, O. Levy & S. Curtarolo, *AFLOW-SYM: platform for the complete, automatic and self-consistent symmetry analysis of crystals*, *Acta Cryst. A* **74**, 184–203 (2018). DOI: [10.1107/S2053273318003066](https://doi.org/10.1107/S2053273318003066). [PDF]

2017

10. A. Hever, **C. Oses**, S. Curtarolo, O. Levy & A. Natan, *The structure and composition statistics of 6A binary and ternary structures*, *Inorg. Chem.* **57**(2), 653–667 (2017). DOI: [10.1021/acs.inorgchem.7b02462](https://doi.org/10.1021/acs.inorgchem.7b02462). [PDF]
9. F. Rose, C. Toher, E. Gossett, **C. Oses**, M. Buongiorno Nardelli, M. Fornari & S. Curtarolo, *AFLUX: The LUX materials search API for the AFLOW data repositories*, *Comput. Mater. Sci.* **137**, 362–370 (2017). DOI: [10.1016/j.commatsci.2017.04.036](https://doi.org/10.1016/j.commatsci.2017.04.036). [PDF]
 - This paper was selected for **Editor's Choice** by Elsevier (2017).
8. O. Isayev[†], **C. Oses**[†], C. Toher, E. Gossett, S. Curtarolo & A. Tropsha, *Universal Fragment Descriptors for Predicting Properties of Inorganic Crystals*, *Nat. Commun.* **8**, 15679 (2017). DOI: [10.1038/ncomms15679](https://doi.org/10.1038/ncomms15679). [PDF]
7. C. Toher, **C. Oses**, J. J. Plata, D. Hicks, F. Rose, O. Levy, M. de Jong, M. Asta, M. Fornari, M. Buongiorno Nardelli & S. Curtarolo, *Combining the AFLOW GIBBS and elastic libraries to efficiently and robustly screening thermomechanical properties of solids*, *Phys. Rev. Mater.* **1**, 015401 (2017). DOI: [10.1103/PhysRevMaterials.1.015401](https://doi.org/10.1103/PhysRevMaterials.1.015401). [PDF]
6. C. Nyshadham, **C. Oses**, J. E. Hansen, I. Takeuchi, S. Curtarolo & G. L. W. Hart, *A Computational High-Throughput Search for New Ternary Superalloys*, *Acta Mater.* **122**, 438–447 (2017). DOI: [10.1016/j.actamat.2016.09.017](https://doi.org/10.1016/j.actamat.2016.09.017). [PDF]
5. S. Sanvito, **C. Oses**, J. Xue, A. Tiwari, M. Žic, T. Archer, P. Tozman, M. Venkatesan, J. M. D. Coey & S. Curtarolo, *Accelerated Discovery of New Magnets in the Heusler Alloy Family*, *Sci. Adv.* **3**(4), e1602241 (2017). DOI: [10.1126/sciadv.1602241](https://doi.org/10.1126/sciadv.1602241). [PDF]

2016

4. A. van Roekeghem, J. Carrete, **C. Oses**, S. Curtarolo & N. Mingo, *High-Throughput Computation of Thermal Conductivity of High-Temperature Solid Phases: The Case of Oxide and Fluoride Perovskites*, *Phys. Rev. X* **6**(4), 041061 (2016). DOI: [10.1103/PhysRevX.6.041061](https://doi.org/10.1103/PhysRevX.6.041061). [PDF]
3. K. Yang, **C. Oses** & S. Curtarolo, *Modeling Off-Stoichiometry Materials with a High-Throughput Ab-Initio Approach*, *Chem. Mater.* **28**(18), 6484–6492 (2016). DOI: [10.1021/acs.chemmater.6b01449](https://doi.org/10.1021/acs.chemmater.6b01449). [PDF]

2015

2. C. E. Calderon, J. J. Plata, C. Toher, **C. Oses**, O. Levy, M. Fornari, A. Natan, M. J. Mehl, G. L. W. Hart, M. Buongiorno Nardelli & S. Curtarolo, *The AFLOW Standard for High-Throughput Materials Science Calculations*, *Comput. Mater. Sci.* **108A**, 233–238 (2015). DOI: [10.1016/j.commatsci.2015.07.019](https://doi.org/10.1016/j.commatsci.2015.07.019). [PDF]
 - This paper was selected for **Editor's Choice** by Elsevier (2015).
1. O. Isayev, D. Fourches, E. N. Muratov, **C. Oses**, K. M. Rasch, A. Tropsha & S. Curtarolo, *Materials Cartography: Representing and Mining Materials Space Using Structural and Electronic Fingerprints*, *Chem. Mater.* **27**(3), 735–743 (2015). DOI: [10.1021/cm503507h](https://doi.org/10.1021/cm503507h). [PDF]
 - This paper was one of the **top 10 most highly downloaded papers** for the month of January 2015 by the American Chemical Society (2015).
 - This paper was selected for **Editors' Choice** by the American Chemical Society (2015).

Book Publications

2019

3. C. Toher, **C. Oses** & S. Curtarolo, *Automated computation of materials properties*, *Materials Informatics: Methods, Tools and Applications*, Ch. 7. DOI: [10.1002/9783527802265.ch7](https://doi.org/10.1002/9783527802265.ch7). [PDF]

2018

2. S. Sanvito, M. Žic, J. Nelson, T. Archer, **C. Oses** & S. Curtarolo, *Machine learning and high-throughput approaches to magnetism*, *Handbook of Materials Modeling. Volume 2 Applications: Current and Emerging Materials*. DOI: [10.1007/978-3-319-50257-1_108-1](https://doi.org/10.1007/978-3-319-50257-1_108-1). [PDF]
1. C. Toher, **C. Oses**, D. Hicks, E. Gossett, F. Rose, P. Nath, D. Usanmaz, D. C. Ford, E. Perim, C. E. Calderon, J. J. Plata, Y. Lederer, M. Jahnátek, W. Setyawan, S. Wang, J. Xue, K. M. Rasch, R. V. Chepurskii, R. H. Taylor, G. Gomez, H. Shi, A. R. Supka, R. Al Rahal Al Orabi, P. Gopal, F. T. Cerasoli, L. Liyanage, H. Wang, I. Siloi, L. A. Agapito, C. Nyshadham, G. L. W. Hart, J. Carrete, F. Legrain, N. Mingo, E. Zurek, O. Isayev, A. Tropsha, S. Sanvito, R. M. Hanson, I. Takeuchi, M. J. Mehl, A. N. Kolmogorov, K. Yang, P. D'Amico, A. Calzolari, M. Costa, R. De Gennaro, M. Buongiorno Nardelli, M. Fornari, O. Levy & S. Curtarolo, *The AFLOW Fleet for Materials Discovery*, *Handbook of Materials Modeling. Volume 1 Methods: Theory and Modeling*. DOI: [10.1007/978-3-319-42913-7_63-2](https://doi.org/10.1007/978-3-319-42913-7_63-2). [PDF]

Talks/Presentations

Powering Tomorrow with AI-Driven High-Entropy Design

61. **Invited talk** for the Symposium on “High-Entropy Ceramics” at the CIMTEC International Conferences on Modern Materials and Technologies 2026; 16th International Ceramics Congress, Perugia, Italy — Jun 21, 2026.
60. **Invited talk** for the Session on “Computational Materials” at the GE Vernova’s Material Frontier Symposium: Powering the Future, Niskayuna, NY — Jun 03, 2026.
59. **Invited talk** for the Symposium on “Engineering Complexity—High Entropy Ceramics as a Design Platform for Next-Generation Materials” at the MRS 2026 Spring Meeting & Exhibit, Honolulu, HI — May 01, 2026.
58. **Invited talk** for the Material Sciences Strategy Offsite for the Johns Hopkins University Applied Physics Laboratory Section R1A, Washington, DC — Mar 26, 2026.
57. **Invited seminar** for the Departmental Seminar Series at the Department of Materials Science and Engineering at the University of Maryland, College Park, MD — Mar 25, 2026.

The Intersection of Energy, Entropy, and Exploration: Data-Driven Discovery of High-Entropy Materials

56. **Invited talk** for the Session on “Data-Driven Thin Film Design: High-Throughput Experimentation, Simulation, and Machine Learning” at the International Conference on Metallurgical Coatings and Thin Films (ICMCTF) 2026, San Diego, CA — Apr 20, 2026.

From Data to Discovery: Active Learning Unlocks Complex Ceramic Design Spaces

55. **Invited talk** for the Symposium on “Advances in Ceramic Materials and Processing” at the TMS 2026 Annual Meeting & Exhibition, San Diego, CA — Mar 16, 2026.
54. **Invited talk** for the Mini Symposium on “Computational Thermodynamics: Energy and Energy Landscape” at the 2025 SIAM New York-New Jersey-Pennsylvania Section Conference, University Park, PA — Nov 02, 2025.

AI Materials Discovery for Nuclear Waste and Spent Fuel Immobilization

53. **Invited talk** for the 2025 Annual Fission Workshop of the Advanced Research Projects Agency-Energy (ARPA-E), Arlington, VA — Oct 02, 2025.

Accelerating Energy Solutions with High-Entropy Materials: Leveraging Disorder, Computation, and AI

52. **Invited talk** for the The Advanced Materials Show at MS&T25 Technical Meeting and Exhibition, Columbus, OH — Sep 30, 2025.

High-Entropy Alloys and Halides: Expanding the Energy-Materials Space

51. **Invited talk** for the Symposium on “Advances in Refractory High Entropy Alloys and Ceramics” at the MS&T25 Technical Meeting and Exhibition, Columbus, OH — Sep 30, 2025.

High-Entropy Oxides and Halides: Expanding the Energy-Materials Space

50. **Invited talk** for the Artificial Intelligence for Materials Science (AIMS) Workshop at the National Institute of Standards and Technology (NIST), Rockville, MD — Jun 10, 2025.
 - *Artificial Intelligence for Materials Science (AIMS) Workshop* recording: <https://www.nist.gov/news-events/events/2025/07/artificial-intelligence-materials-science-aims-workshop>

High-Entropy Halides: Expanding the Energy-Materials Space

49. **Invited talk** for the Symposium on “High Entropy and Complex Structure for Electrocatalysis and Other Applications” at the ACS Spring 2025 Meeting & Expo, San Diego, CA — Mar 25, 2025.
48. **Invited talk** for the Symposium on “Advances in Ceramic Materials and Processing” at the TMS 2025 Annual Meeting & Exhibition, Las Vegas, NV — Mar 24, 2025.

Metal Iodide Materials for Energy Applications

47. **Invited talk** for the Symposium on “Understanding High Entropy Materials Via Data-Science and Computational Approaches” at the MS&T24 Technical Meeting and Exhibition, Pittsburgh, PA — Oct 09, 2024.

Success Stories in Computationally-Driven Materials Discovery

46. **Invited seminar** for the Departmental Seminar Series at the Department of Materials Science and Engineering at the University of Connecticut, Storrs, CT — Sep 27, 2024.
45. **Contributed talk** for the Symposium on “Machine learning assisted materials discovery” at the 11th International Conference on Multiscale Materials Modeling (MMM11), Prague, Czech Republic — Sep 24, 2024.
44. **Invited seminar** for the Departmental Seminar Series at the Department of Chemical and Nano Engineering at the University of California, San Diego, San Diego, CA — May 01, 2024.
43. **Invited seminar** for the Materials Science and Engineering Fall Seminar Series at the Rensselaer Polytechnic Institute, Troy, NY — Oct 18, 2023.
42. **Invited seminar** for the Physics Department Colloquium at Georgetown University, Washington, DC — Sep 26, 2023.
41. **Invited seminar** for the OneChemistry Symposium, Johns Hopkins University — Apr 18, 2023.
40. **Invited seminar** for the Computational Spintronics Group, Trinity College Dublin, Ireland — Feb 17, 2023.
39. **Invited seminar** for the Physics Department Colloquium at Johns Hopkins University, Baltimore, Maryland — Feb 15, 2023.

Disorder by design: Applications and modeling of high-entropy ceramics

38. **Invited talk** for the Symposium on “Advancing Ab-Initio Force Fields with Machine-Learning for Energy Materials” at the International Conference on Computational & Experimental Engineering and Sciences (ICCES 2024), Singapore — Aug 04, 2024.
37. **Invited seminar** for the Departmental Seminar Series at the University of Michigan, Ann Arbor, MI — Mar 04, 2024.
36. **Invited seminar** for the Johns Hopkins University Applied Physics Laboratory, Baltimore, Maryland — Mar 09, 2023.
35. **Invited seminar** for the Hopkins Extreme Materials Institute at Johns Hopkins University, Baltimore, Maryland — Nov 01, 2022.
34. **Invited seminar** for the Department of Materials Science and Engineering at Texas A&M University, College Station, Texas — Feb 10, 2022.

Bayesian Optimization of the PhD (and beyond)

33. **Keynote** for the Fifth Annual Research Summit for the University Center of Exemplary Mentoring at Duke University, Durham, NC — Feb 29, 2024.

Computational Materials Science

32. **Invited seminar** for the PARADIM Summer School, Johns Hopkins University — Aug 02, 2023.

Formation Descriptors for High-Entropy High-Hardness Metal Carbides

31. **Invited talk** for the Machine Learning in Ceramics and Glasses Webinar at the Institute of Materials, Minerals and Mining (IOM3), London, UK — Mar 29, 2022.

High-entropy ceramics

30. **Invited seminar** for the Department of Materials Science and Engineering at Michigan Technological University, Houghton, Michigan — Mar 17, 2022.
29. **Invited seminar** for the Department of Materials Science and Engineering at the University of California, Irvine, Irvine, CA — Mar 04, 2022.
28. **Invited seminar** for the Department of Physics at the University of Alabama at Birmingham, Birmingham, Alabama — Feb 04, 2022.
27. **Invited seminar** for the Department of Mechanical Engineering at Rowan University, Glassboro, NJ — Jan 27, 2022.
26. **Invited seminar** for the Department of Materials Science and Engineering at Johns Hopkins University, Baltimore, MD — Jan 04, 2022.
25. **Invited seminar** for the Department of Mechanical Engineering at Texas A&M University, College Station, Texas — Feb 24, 2021.
24. **Invited seminar** for the Lecture Series in Materials Science & Engineering at the North Carolina State University, Raleigh, North Carolina — Jan 22, 2021.

Data for Materials Development Platforms

23. **Invited seminar** for the aiM Program Boot Camp and Orientation at Duke University, Durham, North Carolina — Aug 19, 2021.
 - *Data for Materials Development Platforms* recording: <https://youtu.be/wLegemRIMpk>

Entropy and ceramics: A valuable partnership

22. **Invited seminar** for the Department of Materials and Interfaces at the Weizmann Institute of Science, Rehovot, Israel — Feb 06, 2020.
21. **Invited seminar** for the Sackler Center for Computational Molecular and Materials Science at Tel Aviv University, Tel Aviv, Israel — Feb 05, 2020.
20. **Invited seminar** for the Department of Materials Engineering at Ben-Gurion University of the Negev, Beer Sheva, Israel — Jan 29, 2020.

Cloud-oriented computational phase diagrams with AFLOW-CHULL

19. **Contributed talk** for the March Meeting of the American Physical Society, Boston, Massachusetts — Mar 07, 2019.
18. **Poster presentation** for the Open Databases Integration for Materials Design (OPTiMaDe) Workshop of CECAM (Centre Européen de Calcul Atomique et Moléculaire), Lausanne, Switzerland — Jun 11, 2018.

Going Off-Stoichiometry: Challenging Traditional Materials Discovery

17. **Invited seminar** for the Center for Computational Materials Science at the Naval Research Laboratory, Washington, DC — Jan 09, 2019.

Universal Fragment Descriptors for Predicting Properties of Inorganic Crystals

16. **Contributed talk** for the 13th World Congress in Computational Mechanics (WCCM) of the International Association for Computational Mechanics (IACM), New York City, New York — Jul 23, 2018.
15. **Contributed talk** for the Mach Conference of the Hopkins Extreme Materials Institute (HEMI), Annapolis, Maryland — Apr 05, 2018.
14. **Contributed talk** for the Chemistry Department's Third Annual Graduate Research Symposium at Duke University, Durham, North Carolina — Oct 09, 2017.
13. **Contributed talk** for the March Meeting of the American Physical Society, New Orleans, Louisiana — Mar 14, 2017.

Advancements in Materials Informatics with AFLOW

12. **Invited seminar** for the Theory Department at the Fritz-Haber-Institut der Max-Planck-Gesellschaft, Berlin, Germany — Jan 18, 2018.
11. **Invited seminar** for the Physics Department at the Humboldt University of Berlin, Berlin, Germany — Jan 16, 2018.

Modeling Off-Stoichiometric Materials with a High-Throughput, Ab-Initio Approach

10. **Contributed talk** for the March Meeting of the American Physical Society, Baltimore, Maryland — Mar 16, 2016.

Materials Cartography: Representing and Mining Materials Space using Structural and Electronic Fingerprints

9. **Invited seminar** for the Condensed Matter Physics Seminar Series at Brigham Young University, Provo, Utah — Feb 18, 2016.
8. **Contributed talk** for the Mechanical Engineering and Materials Science (MEMS) Department's Graduate Student Seminar Series at Duke University, Durham, North Carolina — Sep 25, 2015.
7. **Contributed talk** for the March Meeting of the American Physical Society, San Antonio, Texas — Mar 02, 2015.

Plume Propagation Simulation for Pulsed Laser Deposition

6. **Poster presentation** for the Machine Learning Summer School (MLSS) at the University of Texas at Austin, Austin, Texas — Jan 12, 2015.
 5. **Contributed talk** for the NSF/AAAS/EHR Emerging Researchers National Conference, Washington, DC — Feb 22, 2014.
 4. **Poster presentation** for the MRS/ASM/AVS/AReMS Meeting, North Carolina State University, Raleigh, North Carolina — Nov 15, 2013.
 3. **Poster presentation** for the Mechanical Engineering and Materials Science (MEMS) Department's Annual Retreat at Duke University, Durham, North Carolina — Aug 22, 2013.
- **Best Presentation Award**

Synchrotron Radiation Focusing Optics — Capillary Beam Stop Design

2. **Contributed talk** for the NSF/AAAS/EHR Emerging Researchers National Conference, Washington, DC — Mar 02, 2013.
- **First Place in Nanoscience and Physics Research Presentation**
1. **Poster presentation** for the LSAMP Research Symposium at Cornell University, Ithaca, New York — Aug 07, 2012.

Teaching Experience

Instructor	Spring 2023–2026	EN.500.113: <i>Gateway Computing: Python</i> , Johns Hopkins University
Instructor	Fall 2023–2025	EN.510.666: <i>Introduction to Computational Materials Modeling</i> , Johns Hopkins University
Co-Instructor	Spring 2021	ME 555: <i>Applications of Artificial Intelligence in Materials</i> , Duke University Department of Mechanical Engineering and Materials Science
Teaching Assistant	Spring 2020	ME 555: <i>Computational Materials Science by Examples and Applications</i> , Duke University Department of Mechanical Engineering and Materials Science

Teaching Assistant Fall 2014–Spring 2015 ME 221: *Structure and Properties of Solids*, Duke University
 Department of Mechanical Engineering and Materials Science
 • **Best Teaching Assistant Award**, Aug 14, 2015

Service [†] contributed equally

Mini Symposium on “Computational Thermodynamics: Energy and Energy Landscape”

Co-Organizers: Z.-K. Liu, J. Deng, T. R. Sinno & R. Wentzcovitch

23. **Co-Organizer and Presenter** at the 2025 SIAM New York-New Jersey-Pennsylvania Section Conference, University Park, PA — Oct 31–Nov 2, 2025.

Johns Hopkins Data Science and AI Institute Fall 2025 Symposium: AI in Daily Life

Co-Organizers: D. J. H. Mathews[†], S. Vedula[†], C. Douville[†], T. Shu[†], K. Vaught[†], B. Ranno, S. Pagano & S. McGhee

22. **Organizer** for the Fall 2025 Symposium at the Johns Hopkins Data Science and AI Institute, Baltimore, MD — Sep 16, 2025.

Data-Driven Materials Modeling Workshop

Co-Organizers: B. Bukowski & T. Curk

21. **Organizer and Presenter** at Johns Hopkins University, Baltimore, Maryland — May 29–31, 2024.

- *Data-Driven Thermodynamic Modeling for Materials Discovery* recording: <https://youtu.be/kZj3zQkBAKg>

Foundations to Futures: Materials Data and AI

Co-Chairs: D. Audus & F. Sen

20. **Conference Co-Chair** at the Materials Research Data Alliance (MaRDA) 2024 Annual Meeting, Baltimore, Maryland — Feb 20–22, 2024.

Focus Session: Computational Design, Understanding and Discovery of Novel Materials

Co-Chairs: E. Jankowski, R. Sundararaman & D. Usanmaz

19. **Session Chair** for the March Meeting of the American Physical Society, Minneapolis, Minnesota — Mar 3–8, 2024.

AI, Data Science — Developing the Role for Sustainable Energy in Hopkins’ Expansion and Vision

Co-Chair: P. Clancy

18. **Session Co-Chair** at the ROSEI 2024 Summit, Baltimore, Maryland — Jan 17, 2024.

AFLOW School: Integrated infrastructure for computational materials discovery

Co-Organizers: C. Toher, D. Hicks, M. Esters, R. Friedrich, E. Gossett, A. Smolyanyuk, H. Eckert, S. Divilov, F. Rose, M. J. Brenner & S. Curtarolo

17. **Presenter** for the Machine Learning for Materials Research Bootcamp of the University of Maryland/NIST/MRS, College Park, Maryland — Aug 10, 2023.
16. **Organizer and Presenter** at Johns Hopkins University, Baltimore, Maryland — Sep 21, 2022.
 - *Introduction and AFLOW-ML: Machine Learning* recording: <https://youtu.be/Xj5BGuFC9ew>
15. **Presenter** for the Machine Learning for Materials Research Bootcamp of the University of Maryland/NIST/MRS, College Park, Maryland — Aug 11, 2022.
14. **Co-Organizer and Presenter** at the East African Institute for Fundamental Research, University of Rwanda, Kigali, Rwanda — Feb 21–24, 2022.
13. **Co-Organizer and Presenter** at the Technische Universität (TU) Dresden and Helmholtz-Zentrum Dresden-Rossendorf — Sep 6–10, 2021.
 - *Introduction to Density Functional Theory and VASP* recording: https://youtu.be/_RsQH3TY7kl
 - *AFLOW-CHULL: Thermodynamics* recording: <https://youtu.be/zcY7gTZIB-Y>
 - *AFLOW-POCC: Disorder* recording: <https://youtu.be/lcDSYiF4AS4>
12. **Co-Organizer and Presenter** at the University of Virginia, Charlottesville, Virginia — Aug 17, 2021.
 - *AFLOW-CHULL and AFLOW-CCE: Thermodynamics* recording: <https://youtu.be/cLhOcN1sQ7M>
11. **Presenter** for the Machine Learning for Materials Research Bootcamp of the University of Maryland/NIST, College Park, Maryland — Jul 29, 2021.
 - *AFLOW-ML: Machine Learning* recording: <https://youtu.be/uFQ-lyTaxCc>
10. **Co-Organizer and Presenter** at Texas A&M University, College Station, Texas — Jul 12–15, 2021.
 - *Introduction to Density Functional Theory and VASP* recording: <https://youtu.be/KXnJGdVgosa>
 - *AFLOW-CHULL and AFLOW-CCE: Thermodynamics* recording: <https://youtu.be/ElaniAcrbhU>
 - *AFLOW-POCC: Disorder* recording: https://youtu.be/D_cfHIlpBiA
9. **Session Chair** for the Virtual Spring Meeting of the Materials Research Society — Apr 17, 2021.
8. **Presenter** for the Materials 4.0 Summer School 2020 at the Dresden Center for Computational Materials Science (DCMS), Technische Universität (TU) Dresden — Aug 18, 2020.
 - *AFLOW-CHULL: Thermodynamics* recording: <https://youtu.be/ncm356YNBVc>
7. **Presenter** for the Machine Learning for Materials Research Bootcamp & Workshop on Machine Learning Microscopy Data of the University of Maryland/NIST, College Park, Maryland — Jul 23, 2020.
 - *AFLOW-ML: Machine Learning* recording: <https://youtu.be/x2qeBtOXues>
6. **Co-Organizer and Presenter** at Texas A&M University, College Station, Texas — Jun 16–18, 2020.
 - *Introduction to Density Functional Theory and VASP* recording: <https://youtu.be/ChySAfo2w7g>
 - *AFLOW-CHULL: Thermodynamics* recording: <https://youtu.be/9Sa8D4inJ5w>
 - *AFLOW-POCC: Disorder* recording: <https://youtu.be/xr-mU-1ShQQ>
5. **Presenter** for the Machine Learning for Materials Research Bootcamp & Workshop on Autonomous Materials Research of the University of Maryland/NIST, College Park, Maryland — Aug 05, 2019.
4. **Co-Organizer and Presenter** at the University of Pennsylvania, Philadelphia, Pennsylvania — May 03, 2019.
3. **Co-Organizer and Presenter** at the North Carolina State University, Raleigh, North Carolina — Mar 12, 2019.
2. **Co-Organizer and Presenter** at Carnegie Mellon University, Pittsburgh, Pennsylvania — Jan 21, 2019.
1. **Presenter** for the Machine Learning for Materials Research Bootcamp & Workshop on Machine Learning Quantum Materials of the University of Maryland/NIST/Moore Foundation, Institute for Bioscience & Biotechnology Research in Gaithersburg, Maryland — Aug 02, 2018.

Press and News Releases

JHU Energy Institute News	Jun 23, 2026	<i>Laboratory to Launchpad Helps Researchers Navigate the Path from Discovery to Impact</i> energyinstitute.jhu.edu/laboratory-to-launchpad-helps-researchers-navigate-the-path-from-discovery-to-impact
JHU Energy Institute News	Feb 05, 2026	<i>Corey Oses Awarded ARPA-E IGNIITE Funding for AI-Driven Fusion Materials Research</i> energyinstitute.jhu.edu/corey-oses-awarded-arpa-e-igniite-funding-for-ai-driven-fusion-materials-research

JHU Energy Institute News	Dec 17, 2025	<i>The Platinum Problem</i> • This press release is featured on JHU Engineering News . energyinstitute.jhu.edu/the-platinum-problem
JHU Engineering News	Jun 16, 2025	<i>Oses and Team Win 2025 Nexus Award</i> engineering.jhu.edu/materials/news/oses-and-team-win-2025-nexus-award
JHU Energy Institute News	Oct 16, 2024	<i>Corey Oses Earns ISMM Early-Career Investigator Award</i> • This press release is featured on JHU Engineering News . energyinstitute.jhu.edu/corey-oses-earns-ismm-early-career-investigator-award
JHU Energy Institute News	Jun 12, 2024	<i>Team Co-Led by Corey Oses Receives SURPASS Award</i> • This press release is featured on JHU Engineering News . energyinstitute.jhu.edu/team-co-led-by-corey-oses-receives-surpass-award
JHU Engineering News	Jun 21, 2023	<i>Unlocking the potential of thermoelectric materials</i> engineering.jhu.edu/news/unlocking-the-potential-of-thermoelectric-materials
Duke Engineering News	Oct 11, 2022	<i>Heat-Proof Chaotic Carbides Could Revolutionize Aerospace Technology</i> pratt.duke.edu/about/news/heat-proof-chaotic-carbides-could-revolutionize-aerospace-technology
White House Office of Science & Technology Policy	Nov 18, 2021	<i>Featured Vignette in the November 2021 Materials Genome Initiative Strategic Plan (page 9)</i> mgi.gov/sites/default/files/documents/MGI-2021-Strategic-Plan.pdf
University of Buffalo	Sep 2019	<i>Scientists predict new forms of superhard carbon</i> • This press release is featured on Phys.org , ScienceDaily , SciTechDaily , and Tribonet . buffalo.edu/ubnow/stories/2019/09/zurek-superhard-carbon.html
Duke Engineering News	Nov 2018	<i>Disordered Materials Could Be Hardest, Most Heat-Tolerant Ever</i> • This press release is featured on AAAS EurekAlert! , Phys.org , ScienceDaily , Science Bulletin , Naaju , NewsBeezer , RemoNews , Tech2 , and LongRoom News . pratt.duke.edu/about/news/chaotic-carbides
MRS Bulletin	Aug 2017	<i>Universal fragment descriptor predicts materials properties</i> cambridge.org/core/journals/mrs-bulletin/news/universal-fragment-descriptor-predicts-materials-properties
UNC Eshelman School of Pharmacy	Jun 2017	<i>Breakthrough Tool Predicts Properties of Theoretical Materials, Finds New Uses for Current Ones</i> • This press release is featured on AAAS EurekAlert! , Phys.org , and ScienceDaily . pharmacy.unc.edu/news/2017/06/06/breakthrough-tool-predicts-properties-theoretical-materials-finds-new-uses-current-ones
Duke Engineering News	Apr 2017	<i>Computers Create Recipe for Two New Magnetic Materials</i> • This press release is featured on Phys.org , Slashdot , Hacker News , Reddit , Engadget , Engineering.com , Science Alert , Azo Materials , Next Big Future , Futurism , New Atlas , and International Business Times . pratt.duke.edu/about/news/predicting-magnets
MRS Bulletin	Apr 2015	<i>Materials fingerprints identified for informatics</i> doi.org/10.1557/mrs.2015.76
Computational Chemistry Highlights	Jan 2015	<i>Materials Cartography: Representing and Mining Materials Space Using Structural and Electronic Fingerprints</i> compchemhighlights.org/2015/01/materials-cartography-representing-and.html
Duke Research	Jan 2015	<i>Molecular Tornado</i> research.duke.edu/molecular-tornado

Duke Graduate School	Oct 2014	<i>Competing for NSF Fellowships: Advice from a Current Fellow</i> gradschool.duke.edu/professional-development/blog/competing-nsf-fellowships-advice-current-fellow
ERN Conference 2013	Feb 2013	<i>2013 Oral and Poster Presentation Award Winners</i> new.emerging-researchers.org/2013-oral-and-poster-presentation-winners

Honors and Awards

Award	2026	Early Career Award: Inspiring Generations of New Innovators to Impact Technologies in Energy 2025 (IGNITE 2025), Advanced Research Projects Agency-Energy (ARPA-E)
Publication Award	2026	Most Popular Articles Collection, Publication in Mater. Horiz. , Royal Society of Chemistry: Materials Horizons
Award	2024	Early-Career Investigator Award in Materials Modelling, International Society of Materials Modeling
Publication Award	2024	Editors' Highlight, Publication in Nat. Commun. , Springer Nature
Award	2023	Reviewer of the Year, 2022, npj Computational Materials
Publication Award	2022	Editor's Choice, Publication in Comput. Mater. Sci. , Elsevier
Publication Award	Nov 16, 2021	"Hot paper", Publication in Nat. Rev. Mater. , Web of Science (Clarivate Analytics) Published in the past two years and received enough citations in July/August 2021 to place it in the top 0.1% of papers in the academic field of Materials Science
Publication Award	2021	Editors' Highlight, Publication in Nat. Commun. , Springer Nature
Publication Award	2018	Editor's Choice, Publication in Comput. Mater. Sci. , Elsevier
Publication Award	2017	Editor's Choice, Publication in Comput. Mater. Sci. , Elsevier
Award	Aug 14, 2015	Best Teaching Assistant Award (ME 221) , Duke University Department of Mechanical Engineering and Materials Science
Publication Award	2015	Editor's Choice, Publication in Comput. Mater. Sci. , Elsevier
Publication Award	2015	Top 10 most highly downloaded papers for the month of January 2015, Publication in Chem. Mater. , American Chemical Society
Publication Award	2015	Editors' Choice, Publication in Chem. Mater. , American Chemical Society
Fellowship	2013–2016	Graduate Research Fellowship, National Science Foundation
Award	Aug 22, 2013	Best Presentation Award at the MEMS Departmental Retreat , Duke University Department of Mechanical Engineering and Materials Science
Award	Mar 02, 2013	First Place in Nanoscience and Physics Research Presentation , NSF / AAAS / EHR Emerging Researchers National Conference
Scholarship	2011–2013	Shell Incentive Fund Scholarship
Scholarship	2010 & 2011	Xerox Corporation Scholarship
Scholarship	2010 & 2011	Intel Academic Award

Grant	Jun 18, 2010	Cornell University Unmanned Air Systems Team awarded \$1,000 grant, AUVSI Student Unmanned Aerial Systems Competition
Scholarship	Fall 2010	Dean's Honor List, Cornell University College of Engineering
Scholarship	2009–2013	Meinig Family Cornell National Scholars • Awarded by Peter Meinig (Past Chairman of the Board of Trustees at Cornell University)